Metals Market Index MMI

MMi Dashboard

Iron Ore Price Indices

Iron Ore Port Stock (FOT Qingdao)
IOPI62 62% Fe Fines RMB/t

902
-4 -0.44%
Mar 24th, 2023

Iron Ore Port Stock (FOT Qingdao)
IOPI65 65% Fe Fines RMB/t

1014
-4.00 -0.39%
Mar 24th, 2023

Iron Port Stock (FOT Qingdao)
IOPI58 58% Fe Fines RMB/t

815
-1 -0.12%
Mar 24th, 2023

Iron Ore Seaborne (CFR Qingdao)
IOSI62 62% Fe Fines USD/dmt

138.15
-0.02 -0.01%
Mar 24th, 2023

Iron Ore Seaborne (CFR Qingdao)
IOSI65 65% Fe Fines USD/dmt

143.25
0.00 0.00%
Mar 24th, 2023

Iron Ore Port Stock (FOT Qingdao)
IOPLI 62.5% Fe Lump RMB/t

995
-31 -3.02%
Week Ending Mar 24th, 2023

Exchange Traded Contracts

DCE Iron Ore 62% Fines
I2305 (Jan) RMB/t (3pm close)

866.50
12.50 1.46%
Mar 24th, 2023 (3pm close)

SGX Iron Ore (CFR Qingdao) 62% Fe Fines
April 23 USD/dmt

118.70
2.20 1.89%
Mar 24th, 2023 (5.30 pm Print)

SHFE Rebar
RB2305 (Jan) RMB/t

4107
37 0.91%
Mar 24th, 2023 (3pm close)

Freight Rates

C3, Tubarao - Qingdao USD/t

20.18
0.62 3.18%
Mar 23rd, 2023

C5, W. Australia - Qingdao USD/t

8.93
0.38 4.48%
Mar 23rd, 2023

Steel Price

Steel Rebar (China Domestic) RMB/t

4210
-140 -3.22%
Week Ending Mar 24th, 2023

Inventory Levels

Iron Ore Inventory at Chinese Ports (35)
million tonnes

132.50
-0.62 -0.47%
Week Ending Mar 24th, 2023

Steel Inventory in China
million tonnes

17.84
-0.34 -1.89%
Week Ending Mar 24th, 2023

Steel Price

Steel HRC (China Domestic) RMB/t

4300
-60 -1.38%
Week Ending Mar 24th, 2023

IRON ORE PORT STOCK INDEX (IOPI)

Mar 24th, 2023		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [1]						
Index	Fe Content	Price	Change	Change %	MTD	YTD	Low [2]	High [2]	Price	Change	Change %	MTD	YTD	Low [2]	High [2]
IOPI62	62% Fe Fines	902	-4	-0.4%	869	880	858	892	122.96	0.04	0.0%	120.06	120.58	117.93	122.93
IOPI58	58% Fe Fines	815	-1	-0.1%	813	798	761	793	111.93	0.40	0.4%	113.29	110.17	105.29	110.04
IOPI65	65% Fe Fines	1014	-4	-0.4%	981	992	970	1003	138.71	0.11	0.1%	136.05	136.43	133.84	138.73

IRON ORE SEABORNE INDEX (IOSI)

Mar 24th, 2023		CFR Qingdao, USD/dry tonne						
Index	Fe Content	Price	Change	Change %	MTD	YTD	Low [2]	High [2]
IOSI62	62% Fe Fines	138.15	-0.02	-0.01%	110.90	123.07	83.90	159.45
IOSI65	65% Fe Fines	143.25	0.00	0.00%	121.50	139.81	94.45	185.82

MARKET COMMENTARY

DCE iron ore futures rose by 0.29today, the main contract closed at 866.5.The traders' willingness to ship is less.The steel mills are less active to purchase.The overall trading sentiment of the market is negative.PBF at Shandong port deal 865-881 yuan/mt, PBF at Tangshan port deal 895 yuan/mt,decrease 10 yuan/mt. Affected by the afternoon market downturn, the sentiment in the iron ore spot market is relatively pessimistic and continues to consume the sentiment of finished product demand data. Fundamentally, as of March 24, the inventory of 35 ports tracked by SMM totaled 132.5 million tons, down 620000 tons from last week and 14.9 million tons from the same period last year. As of now, the inventory of 35 ports has decreased for four consecutive periods. The daily average port dredging volume of imported mines in this period increased by 6000 tons to 2.914 million tons on a weekly basis. In the future, the rising space of molten iron in the blast furnace continues to narrow. The impact of overseas weather is gradually fading, and the supply and demand gap in the iron ore market may gradually expand. Overall, the current industrial chain has not fully formed a negative feedback logic, and in the short term, iron ore prices are still

IRON ORE PORT LUMP INDEX (IOPLI)

Week Ending Mar 24th, 2023		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [3]						
Index	Fe Content	Price	W-o-W	Change %	MTD	YTD	Low [2]	High [2]	Price	W-o-W	Change %	MTD	YTD	Low [2]	High [2]
IOPLI62	62.5% Fe Lump	995	-31	-3.0%	917	1033	770	1405	128.66	-4.59	-3.45%	118.08	138.23	95.00	199.00

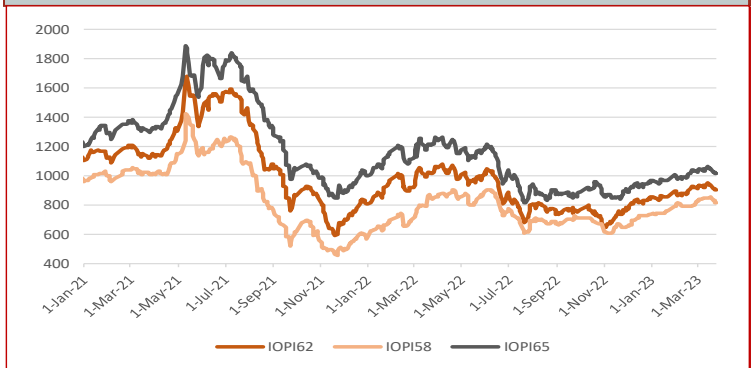
IRON ORE DOMESTIC CONCENTRATE SPOT PRICE ASSESSMENTS AND COMPOSITE INDEX

Week Ending Mar 24th, 2023				RMB/tonne (excluding tax) [3]				USD/tonne (excluding tax) [3]			
Province	Region	Product	Basis	This week	Change %	Low [2]	High [2]	This week	Change %	Low [2]	High [2]
Hebei	Hanxing	66% Fe Concentrate	Dry	1107	2.3%	779	1645	161.25	2.89%	110.31	255.69
Hebei	Qian'an	65% Fe Concentrate	Dry	1085	-0.9%	780	1630	158.05	-0.35%	110.51	251.57
Liaoning	Anshan	65% Fe Concentrate	Wet	845	0.0%	620	1310	123.09	0.57%	87.40	202.32
Shandong	Zibo	65% Fe Concentrate	Dry	1133	2.6%	800	1752	165.04	3.21%	117.19	272.32

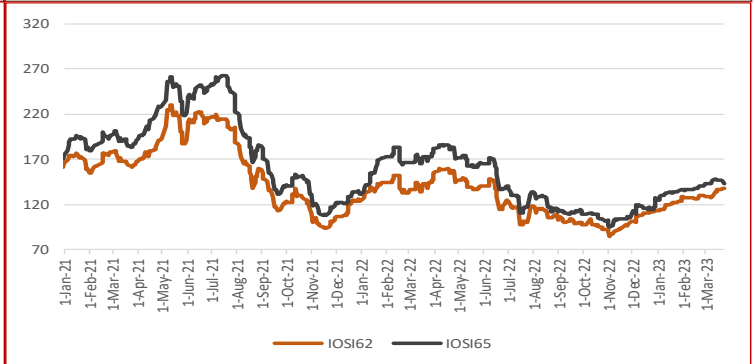
Week Ending Mar 24th, 2023	This week	Change %	Low [2]	High [2]
China Mines Concentrate Composite Index RMB/WT	983.82	-0.06%	706.36	1511.22

[1] Exchange rate applied: RMB/USD = 6.7522 [2] Last 12 months
[3] Weekly exchange rate applied: RMB/USD =6.8651

IRON ORE PORT INDEX, FOT QINGDAO (RMB/WT)



IRON ORE SEABORNE INDEX, CFR QINGDAO (USD/DMT)



IRON ORE PORT STOCK INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Mar 24th, 2023		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne						
Index	Fe Content	November	December	January	February	MTD	QTD	YTD	November	December	January	February	MTD	QTD	YTD
IOPI62	62% Fe Fines	714	822	856	892	869	881	880	92.18	109.43	117.00	121.64	120.06	120.90	120.58
IOPI58	58% Fe Fines	635	711	751	799	813	792	798	82.39	95.17	103.34	109.79	113.29	109.35	110.17
IOPI65	65% Fe Fines	865	935	968	1003	981	993	992	112.37	124.90	132.80	137.36	136.05	136.72	136.43

IRON ORE SEABORNE INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Mar 24th, 2023		CFR Qingdao, USD/dry tonne						
Index	Fe Content	November	December	January	February	MTD	QTD	YTD
IOSI62	62% Fe Fines	93.05	108.92	119.76	127.73	110.90	110.81	123.07
IOSI65	65% Fe Fines	102.40	116.83	132.39	138.06	121.50	121.84	139.81

FREIGHT RATES

Mar 23rd, 2023		FREIGHT RATES - DRY BULK US$/wet tonne				
Route	Designation		Change	Change %	Low [2]	High [2]
W. Australia - Qingdao	C5	8.93	0.38	4.48%	3.57	16.77
Tubarao - Qingdao	C3	20.18	0.62	3.18%	6.70	36.40

IRON ORE PORT LUMP INDEX MONTHLY, QUARTERLY AND YEAR-TO-DATE AVERAGES

Week Ending Mar 24th, 2023		FOT Qingdao (inc. 13% VAT), RMB/wet tonne							CFR Qingdao Equivalent (exc. 13% VAT), USD/dry tonne [1]						
Index	Fe Content	November	December	January	February	MTD	QTD	YTD	November	December	January	February	MTD	QTD	YTD
IOPLI62	62.5% Fe Lump	811	922	964	988	917	915	1033	100.83	117.06	126.47	129.95	118.08	118.39	138.23

IRON ORE INDEX PREMIUMS/DISCOUNTS

Mar 24th, 2023		PORT STOCK INDEX (RMB/WT)	
Index	Fe Content	Spread to IOPI62	% Spread to IOPI62
IOPI58	58% Fe Fines	-87	-9.65%
IOPI65	65% Fe Fines	112	12.42%

Mar 24th, 2023		SEABORNE INDEX (USD/DMT)	
Index	Fe Content	Spread to IOSI62	% Spread to IOSI62
IOSI65	65% Fe Fines	5.10	3.69%

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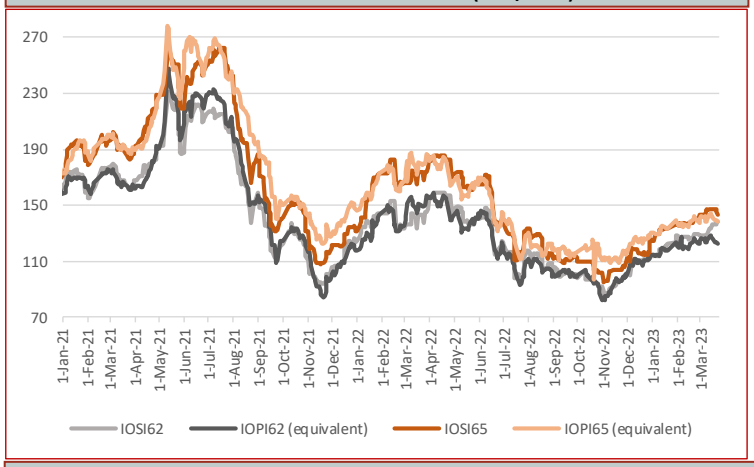
IRON ORE INDEX COMPARISONS (USD/DMT)



CHINA DOMESTIC COMPOSITE MINES INDEX (RMB/Wet Tonne, including VAT)

IRON ORE BRAND SPOT PRICE ASSESMENTS

Mar 24th, 2023	PORT STOCK INDEX (RMB/WT)		
	Price	Change	Diff to IOPI62
Roy Hill	845	-14	-57
SIMEC Fines	775	-14	-127
PB Fines	870	-13	-32
Newman Fines	871	-14	-31
MAC Fines	852	-14	-50
Jimblebar Blended Fines	767	-14	-135
Carajas Fines	1083	-14	181
Brazilian SSF	867	-14	-35
Brazilian Blend Fines	883	-14	-19
RTX Fines	785	-14	-117
West Pilbara Fines	813	-14	-89

Mar 24th, 2023	SEABORNE INDEX (USD/DMT)		
	Price	Change	Diff to IOSI62
Roy Hill	133.65	0.00	-4.50
SIMEC Fines	130.15	0.00	-8.00
PB Fines	134.40	0.00	-3.75
Newman Fines	137.30	0.00	-0.85
MAC Fines	134.40	0.00	-3.75
Jimblebar Blended Fines	126.80	0.00	-11.35
Carajas Fines	167.70	0.00	29.55
Brazilian SSF	141.90	0.00	3.75
Brazilian Blend Fines	143.55	0.00	5.40
RTX Fines	128.05	0.00	-10.10
West Pilbara Fines	132.40	0.00	-5.75

Mar 24th, 2023	PORT STOCK INDEX (RMB/WT)		
	Price	Change	Diff to IOPI58
SSF	815	-1	0
FMG Blended Fines	825	-1	10
Robe River	826	-1	11
Western Fines	829	-1	14
Atlas Fines	822	-1	7
Yandi	807	-1	-8

IRON ORE INDEX NORMALISATION DIFFERENTIALS

Port Stock Index Product Differentials (RMB/wet tonne)

	Applicable range	Value	Change
1% Fe	High Grade Fe 60 - 63%	5.00	0.00
	High Grade Fe 63 - 64%	25.00	-4.00
	High Grade Fe 64 - 65%	25.00	-4.00
	High Grade Fe 65 - 65.5%	25.00	-4.00
	Low Grade Fe	19.00	2.00
1% Alumina	High Fe Grade Al <2.25%	5.00	-5.00
	High Fe Grade Al 2.25-4%	5.00	0.00
	Low Fe Grade Al <2.25%	27.00	-7.00
	Low Fe Grade Al 2.25-4%	40.00	-1.00
1% Silica	High Fe Grade Si <4%	5.00	0.00
	High Fe Grade Si 4-6.5%	5.00	0.00
	Low Fe Grade	6.00	1.00
0.01% Phosphorus	High Fe Grade 0.09%<P<0.115%	5.00	0.00
	High Fe Grade 0.115%<P<0.15%	5.00	0.00
	Low Fe Grade 0.09<P<0.1%	5.00	0.00

Seaborne Index Product Differentials (USD/dry tonne)

	Applicable range	Value	Change
1% Fe	High Grade Fe 60 - 63%	1.75	-0.50
	High Grade Fe 63 - 64%	2.50	0.00
	High Grade Fe 64 - 65%	2.50	0.00
	High Grade Fe 65 - 65.5%	2.50	0.00
1% Alumina	High Fe Grade Al <2.25%	3.00	0.00
	High Fe Grade Al 2.25-4%	2.00	-0.50
1% Silica	High Fe Grade Si <4%	3.00	-0.75
	High Fe Grade Si 4 - 6.5%	1.00	-0.50
0.01% Phosphorus	High Fe Grade 0.09%<P<0.115%	1.25	0.00
	High Fe Grade 0.115%<P<0.15%	5.00	0.00

Port Stock Price Differentials to Qingdao Port for PB Fines (RMB/wet tonne)

Port	Value	Change	Port	Value	Change	Port	Value	Change	Port	Value	Change
Bayuquan	-100.00	0.00	Fangcheng	-25.00	0.00	Lanshan	0.00	0.00	Rizhao	0.00	0.00
Beilun	0.00	0.00	Jiangyin	-30.00	0.00	Lianyungang	0.00	0.00	Shekou	0.00	0.00
Caofeidian	-10.00	0.00	Jingtang	-10.00	0.00	Majishan	0.00	0.00	Taicang	-30.00	0.00
Dalian	0.00	0.00	Lanqiao	0.00	0.00	Qingdao	0.00	0.00	Tianjin	0.00	0.00

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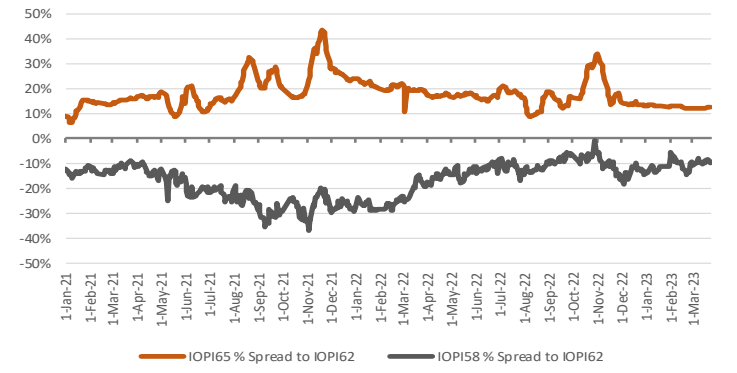
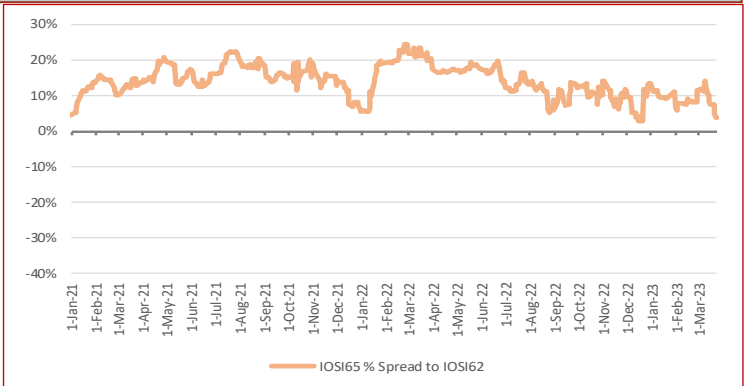
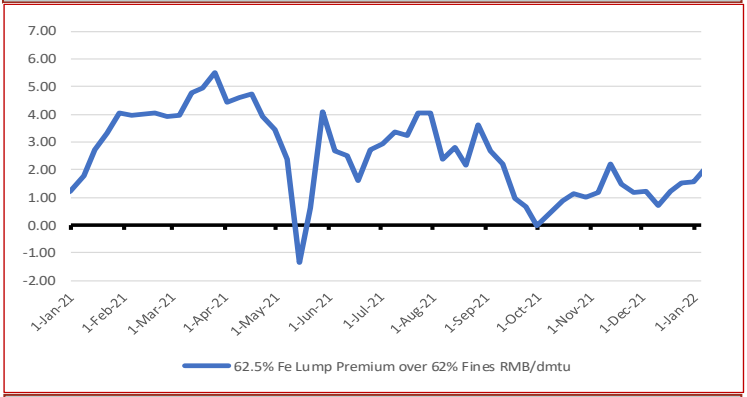
IRON ORE INDEX PREMIUMS/DISCOUNTS

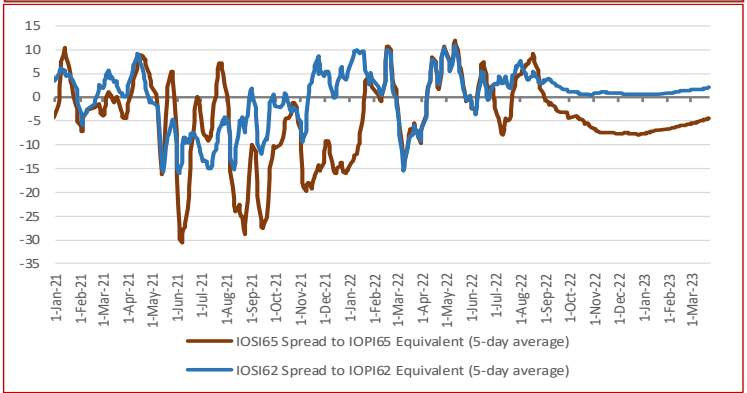




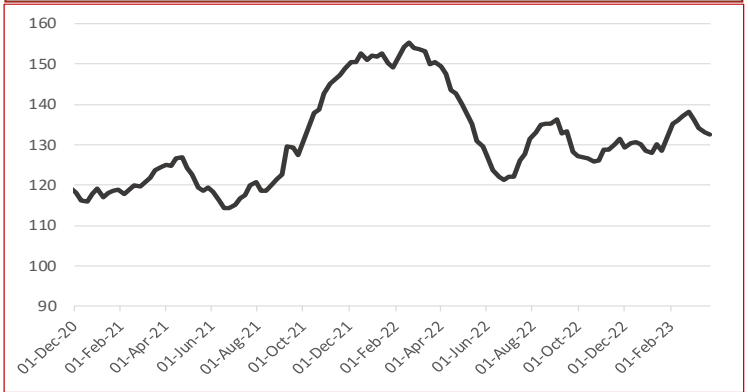
WEEKLY IRON ORE PORT STOCK LUMP PREMIUM (RMB/DMTU)



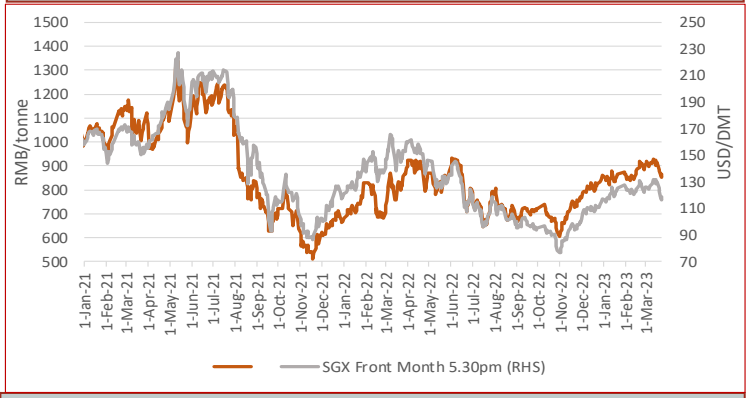
IRON ORE SEABORNE TO PORT STOCK PRICE SPREADS (USD/DMT)



TOTAL IRON ORE INVENTORIES AT CHINA PORTS (MILLION TONNES)



FUTURE TRADING—FRONT MONTH CLOSING PRICE



IRON ORE PORT INVENTORIES (MILLION TONNES)

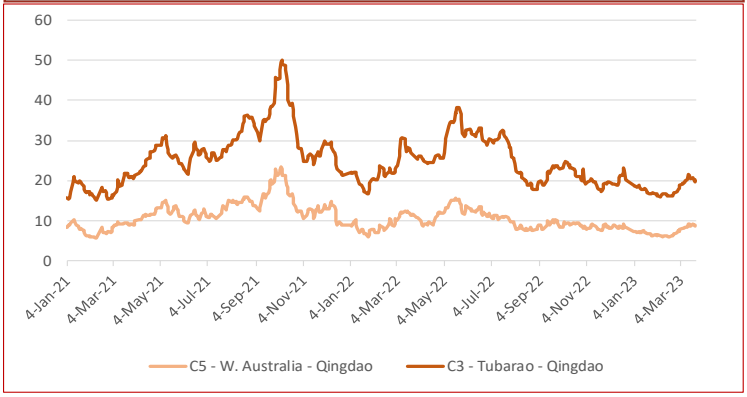
Week Ending Mar 24th, 2023

Province	This week	Change %	Low [2]	High [2]
Jingtang	12.97	-1.59%	9.84	19.20
Qingdao	25.25	0.40%	9.41	26.24
Caofeidian	12.23	4.80%	9.05	16.29
Tianjin	9.91	-2.17%	7.14	12.97
Rizhao	18.93	0.32%	9.44	19.26
Total (35 Ports)	132.50	-0.47%	98.80	155.39

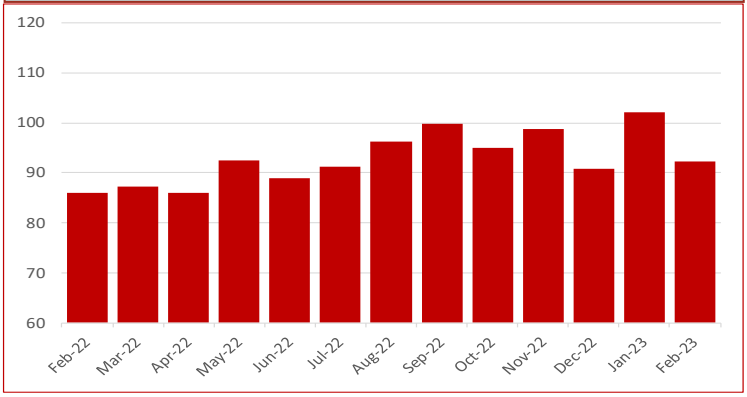
IRON ORE FUTURES CONTRACTS

	DCE (RMB/WMT)			SGX (USD/DMT)		
Closing Date	Mar 24th, 3pm close			Mar 24th, 5:30pm		
Contract	I2305	Change	Change %	Apr. 23	Change	Change %
Closing Price	866.50	12.50	1.46%	118.70	2.20	1.89%
Vol traded ('000 lots)	65.21	1.13	1.76%	5.97	-0.45	-6.96%
Open positions ('000 lots)	58.87	-2.98	-4.81%	27.11	-1.57	-5.46%
Day Low	844.0	-7.00	-0.82%	117.40	-0.50	-0.42%
Day High	871.5	-5.00	-0.57%	120.35	-2.30	-1.88%

DRY BULK FREIGHT RATES (USD/MT)



TOTAL CHINA IRON ORE IMPORT VOLUMES (MILLION TONNES)



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STEEL SPOT MARKET PRICES—CHINA

Steel Spot Market RMB/tonne

Product	2023/3/24	Change	Change %
ReBar HRB400 ϕ18mm	4,210	-140	-3.22%
Wirerod Q300 ϕ6.5mm	4,420	-140	-3.07%
HRC Q235/SS400 5.5mm*1500*C	4,300	-60	-1.38%
CRC SPCC/ST12 1.0mm*1250*2500	4,770	-30	-0.63%
Medium & Heavy Plate Q235B 20mm	4,430	-90	-1.99%
GI ST02Z 1.0mm*1000*C	5,100	-30	-0.58%
Colour Coated Plate	7,500	-50	-0.66%

CHINESE STEEL MILL PROFITABILITY

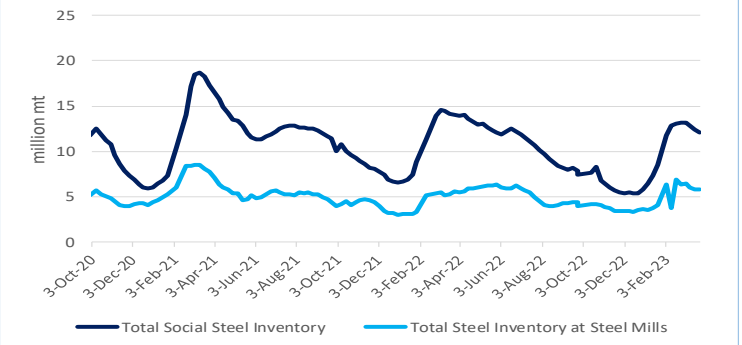
SMM Tracking of Steel Mill P&L - Rebar and Hot-rolled Coil (RMB/tonne)

Category	Price	Change (WoW)	Note
MMi (Fe 62%), USD/mt exluding tax	138.17	2	Mmi CFR Equivalent index for 1st Feb
Coke	2,960	0	2nd grade met coke, Tangshan, incl. tax
Steel Scrap	3,120	-110	steel scrap (6mm) in Zhangjiagang, exl. tax
Billet Cost	3,628	-72	Q234, incl. tax
Rebar cost - Blast furnace	4,133	-81	calculated based on theoretical weight, incl. tax
Rebar profit - Blast furnace	17	-29	based on Shanghai prices, incl. tax
Hot-rolled coil cost - Blast furnace	4,141	-80	based on actual weight, incl. tax
Hot-rolled coil proft - Blast furnace	149	-10	based on Shanghai prices, incl. tax

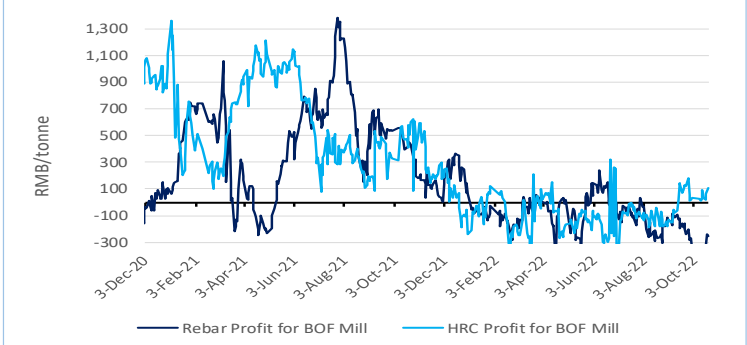
Note: 1. Costs in the table are caluclated based on todays market prices and facout our management, sales, financial and depreciations fe

2. The cost refers to average cost in the industry based on SMM's survey of small, medium and large mills in China

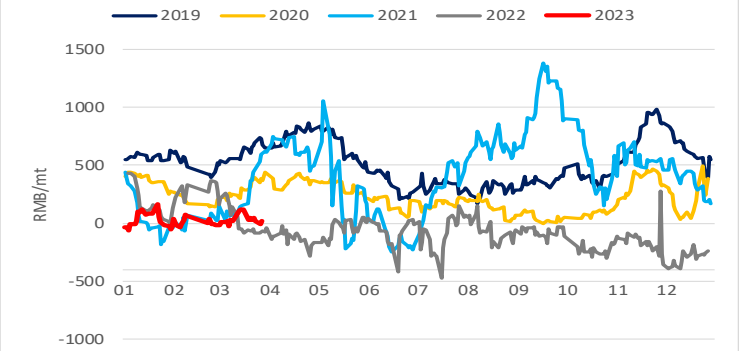
CHINESE STEEL INVENTORIES



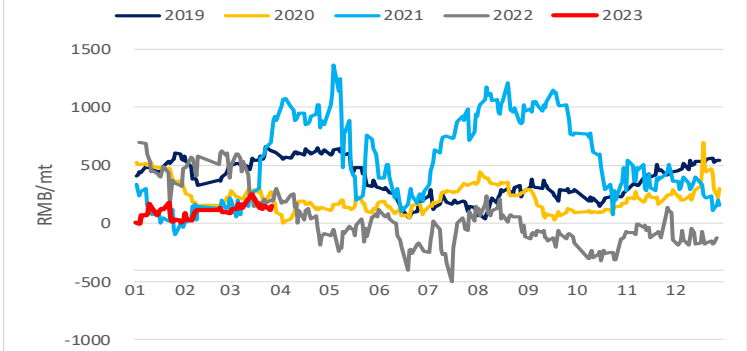
CHINESE STEEL MILL PROFITABILITY



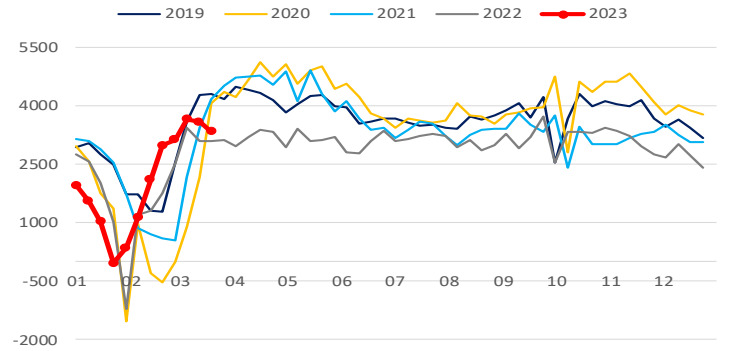
CHINESE STEEL MILL PROFITABILITY—Rebar



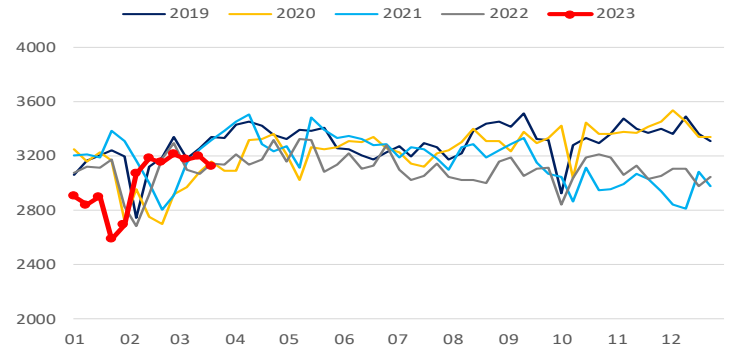
CHINESE STEEL MILL PROFITABILITY—Hot-rolled Coil



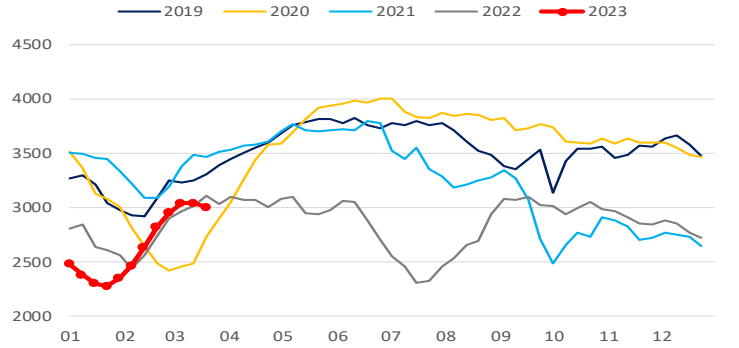
CHINESE STEEL CONSUMPTION—Rebar



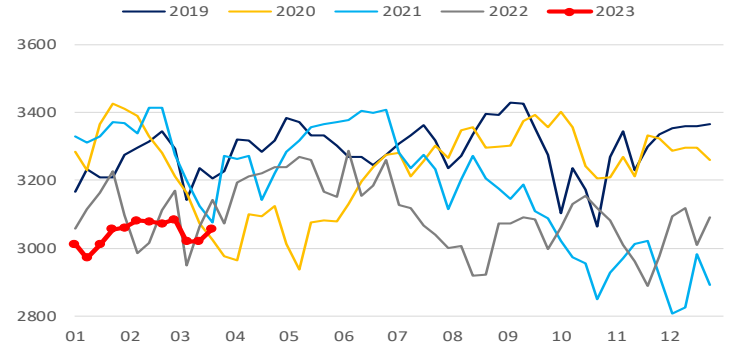
CHINESE STEEL CONSUMPTION—Hot-rolled Coil



CHINESE STEEL PRODUCTION—Rebar



CHINESE STEEL PRODUCTION—Hot-rolled Coil



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IRON ORE INDEX SPECIFICATIONS, COMPILATION RATIONALE AND DATA EXCLUSIONS

Iron Ore Index Specifications (Port and Seaborne)

	65% Fe Fines	62% Fe Fines	58% Fe Fines	62.5% Fe Lump
Fe %	65.00	62.00	58.00	62.50
Alumina %	1.40	2.25	2.25	1.50
Silica%	1.50	4.00	5.50	3.50
Phosphorus %	0.06	0.09	0.05	0.08
Sulphur %	0.01	0.02	0.02	0.02
Moisture %	8.00	8.00	9.00	4.00
Sizing	Granular size below 10mm for at least 90% of cargo; maximum of 40% below 150 micron			Size below 6.3mm max 15% Size above 31.15mm max 25%
Pricing Point	Qingdao Port (FOT and CFR respectively)			FOT Qingdao Port
Timing (Seaborne)	Loading within 4 weeks, Delivery within 8 weeks			Delivery within 2 weeks
Payment Terms	L/C at sight			L/C at sight or CAD

Iron Ore Index Compilation Rationale and Data Exclusions

MMi iron ore indices are compiled from data provided by companies that are part of the iron ore supply chain and involved in spot market transactions. The indices are calculated using detailed transaction-level data submitted to MMi by these companies. This data is normalised to the appropriate specifications and screened to remove outliers before volume-weighted average prices are calculated from the remaining core set of data.

For more details on MMi's iron ore methodology please download the guide published on our website at: www.mmiprices.com

Data Exclusions*

	62%	58%	65%
Port Index	0	0	0
Seaborne index	0		0
Lump Index 62.5	0		

* Number of price submissions for iron ore indices that were excluded from index calculations today as they were anomalous and could not be verified

IRON ORE DOMESTIC CONCENTRATE INDEX CALCULATION METHODOLOGY

The compilation method for price index generally refers to the compilation method of CPI price index and other price indices, breakdown the price data and calculate the average value according to a certain method, taking the vertical axis as the regional composite index (average of different grade index) and the horizontal axis as the grade composite index (average of different regional index), a total composite index for domestic ore can be output ultimately. The process system is also adopted in the calculation i.e. each sub-index can be obtained as well.

AVERAGE IRON ORE SPECIFICATIONS APPLIED FOR BRAND PRICE ASSESSMENTS

PORT STOCK BRANDS

Mar 24th, 2023	Specifications applied for 62% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
Roy Hill	60.88%	2.35%	4.45%	0.055%	9.26%
SIMEC Fines	60.00%	2.30%	6.30%	0.060%	6.00%
PB Fines	61.64%	2.31%	3.87%	0.099%	9.45%
Newman Fines	62.49%	2.31%	3.98%	0.090%	7.72%
MAC Fines	60.43%	2.38%	4.64%	0.078%	7.57%
Jimblebar Blended Fines	60.87%	2.95%	4.53%	0.111%	7.32%
Carajas Fines	64.90%	1.39%	1.82%	0.080%	7.78%
Brazilian SSF	62.00%	1.00%	6.50%	0.040%	6.00%
Brazilian Blend Fines	62.56%	1.58%	4.79%	0.091%	8.72%
RTX Fines	61.00%	3.10%	4.50%	0.135%	7.50%
West Pilbara Fines	60.10%	2.30%	4.70%	0.075%	8.50%

Mar 24th, 2023	Specifications applied for 58% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
SSF	56.49%	3.20%	6.19%	0.065%	9.18%
FMG Blended Fines	58.20%	2.43%	5.54%	0.057%	8.29%
Robe River	56.44%	3.16%	5.73%	0.042%	8.44%
Western Fines	57.88%	2.87%	7.50%	0.062%	7.45%
Atlas Fines	58.00%	1.85%	5.50%	0.090%	9.00%
Yandi	56.87%	1.58%	6.41%	0.042%	9.53%

SEABORNE BRANDS

	Specifications applied for 62% brand assessments				
	Fe	Alumina	Silica	Phos	Moisture
Roy Hill	60.70%	2.30%	4.90%	0.055%	8.00%
SIMEC Fines	60.00%	2.30%	6.30%	0.060%	6.00%
PB Fines 62%	62.00%	2.60%	4.30%	0.090%	9.50%
Newman Fines	62.30%	2.40%	4.30%	0.095%	8.00%
MAC Fines	60.80%	2.35%	4.40%	0.075%	7.80%
Jimblebar Blended Fines	60.50%	3.10%	4.80%	0.120%	7.80%
Carajas Fines	65.10%	1.50%	1.70%	0.080%	8.50%
Brazilian SSF	62.00%	1.00%	6.50%	0.040%	6.00%
Brazilian Blend Fines	63.00%	1.50%	5.00%	0.070%	7.00%
RTX Fines	61.00%	3.10%	4.50%	0.135%	7.50%
West Pilbara Fines	60.10%	2.30%	4.70%	0.075%	8.50%

BLOOMBERG TICKERS

PORT STOCK INDICES

	FOT Qingdao (RMB/wet tonne)	CFR Qingdao Equivalent (USD/dry tonne)
IOPI62	IRCNQ001	IRCNQ004
IOPI58	IRCNQ002	IRCNQ005
IOPI65	IRCNQ003	IRCNQ006
IOPLI62	IRCN0036	IRCN0037

SEABORNE INDICES

	CFR Qingdao (USD/DMT)
IOSI62	IRCN0034
IOSI65	IRCN0035

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